Candidate's Name: ..... Index No. .....

School: ..... Signature: .....

553/1
BIOLOGY
Paper 1
(Theory)
July/August 2026
$2\frac{1}{2}$ hours

For Scorers' Use Only

Total Weighted Score		
Initials		

WAKISSHA JOINT MOCK EXAMINATIONS
Uganda Certificate of Education

INSTRUCTIONS TO CANDIDATES:

This paper consists of **two** *sections;* **A** *and* **B.** *It has* ***seven*** *examination items.*

Sections **A** *has* ***three compulsory*** *items.*

Responses to section **A must** *be written in the spaces provided.*

Section **B** *has* **two** *parts;* **I** *and* **II.** *Respond to* **one** *item from each part.*

Responses to Section **B must** *be written in the answer booklet(s)/sheets provided.*

Responses to each item in Section **B** *should start on a fresh page.*

Respond to **five** *items in all.*

Any additional item(s) responded to will **not** *be scored.*

FOR SCORERS' USE ONLY

Item No.	Basis Code	Weighted Score	Scorers' Initials
1	PU		
	PA		
2	CU		
	CA		
3	HU		
	HA		
__(4/5)	EU		
	EA		
__(6/7)	DU		
	DA		
Total Weighted Score			

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SECTION A

Respond to **all** *the items in this section in the spaces provided.*

Item 1

A study was conducted in a semi-arid savannah ecosystem to investigate how plants survive under extreme environmental stress. They observed that some parts of the study area experienced very high temperatures, intense sunlight, low soil moisture, high rates of evaporation, and sandy soils with low nutrient content, particularly nitrogen.
In this environment, several plant species such as acacias (leguminous plant) were thriving. Some plants had small needle-like leaves, while others had thick fleshy stems. A number of plants showed deep tap-root systems, while others had widely spread shallow roots close to the soil surface. The legumes had reduced leaf surface area and waxy cuticles, they maintained relatively healthy growth despite the harsh conditions.
The scientists also observed that many plants had stomata that remained closed during day but opened at night, and some leaves showed a folded structure during peak heat hours.
The ecologists were interested in how the plant species managed to adapt and compete successfully in such a challenging environment.

Task

(a) Identify the plant structures whose functioning was being affected by the environmental conditions in this semi-arid ecosystem.

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For scorer's use only

Basic code	PU
Score	

(b) Explain how the environmental conditions in the scenario affected the processes of the plants.

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For scorer's use only

Basic code	PU
Score	

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(c) Explain how the acacia plants were able to survive and compete successfully with other plants under these conditions.

For scorer's use only

Basic code	PA
Score	

Item 2

In Lufunda Parish, a rural farming community, the local healthy centre has reported a rising number of chronic health conditions affecting both adults and youth. During a medical outreach program, health workers observed the following:-

- many adults especially those above 45 years, are suffering from weak bones, frequent fractures, and persistent back pain (osteoporosis).
- many women and children have visible swelling at the front of the neck (Goitre).
- several middle- aged individuals were diagnosed with diabetes mellitus.
- many youth frequently consume alcohol, tobacco and other substances.
- some patients also showed reduced sensitivity to environmental stimuli such as light and sound due to prolonged substance abuse.

The medical team confirmed that the conditions were linked to hormonal imbalance, enzyme inefficiency, poor diet and substance abuse affecting their body systems. Clinical tests confirmed diabetes, goiter and osteoporosis.

Task

(a) Explain how hormonal imbalance contributes to the development of diabetes, goiter and osteoporosis in the individuals of Lufunda Parish.

For scorer's use only

Basic code	CU
Score	

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(b) Explain how reduced enzyme activity affected nutrient absorption and overall health of the Lufunda people.

For scorer's use only

Basic code	CU
Score	

(c) Discuss the role of diet and lifestyle in the prevention and management of the disorders identified in the community.

For scorer's use only

Basic code	CA
Score	

Item 3

A couple living with HIV has three children with normal vision, however the fourth child, a boy was found to have difficulty in distinguishing between red and green colours. This caused concern to the parents since neither of them had any noticeable vision problems. After medical examination, the doctor diagnosed the child with red-green colour blindness, a genetic condition linked to the X-chromosomes. The doctor re-assured them that the child could live a normal life with proper support.

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Task

(a) Using genetic symbols show how the condition in their fourth child came about.

For scorer's use only

Basic code	CU
Score	

(b) Explain how the couple can;

(i) Continue living positively with their health status.

For scorer's use only

Basic code	CA
Score	

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(ii) Take care of their fourth born child.

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For scorer's use only

Basic code	CA
Score	

SECTION B

Part 1

Respond to only **one** *item from this part.*

Responses should be written in the answer booklet(s) provided.

Item 4

Residents of Kayanja Village depended on a nearby wetland for clean water, fishing and harvesting papyrus for making mats and baskets. The wetland also supported a wide variety of birds, insects and helped regulate flooding during rainy season. Over time, a real estate developer drained a large portion of the wetland and filled it with sand to construct residential houses. In addition, waste water from houses was directed into a small channel that originally fed the wetland.

After a few years residents began to notice reduced fish populations, frequent flooding during heavy rains, foul smelling water, and increase in water borne diseases.

Task

(a) Explain how the developer's activities lead to the loss of the benefits previously obtained from the wetland by both the environment and the residents.

(b) Suggest the importance of conserving wetland like the one described in the scenario.

Item 5

The Uganda Wild Life Authority conducted a study in Murchison fall National Park and surrounding communities to access the impact of increasing human activities on Wildlife population and ecosystem stability.

Over the past five years there has been; expansion of farmland into protected areas, increased poaching of herbivores such as antelopes, charcoal burning and deforestation near Park boundaries.

At the same time communities have reported rising cases of crop destruction by antelopes, increased attacks on livestock by lions and reduced fish catch from the nearby water bodies.

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The figure below shows changes in wild life population fish and human activities (2016-2020).

Year	Antelope population	Lion population	Poaching Cases	Fish Population
2016	12,000	320	80	70
2017	11,200	310	110	65
2018	10,060	290	150	55
2019	8,800	270	210	40
2020	7,500	212	260	32

The changes in fish population was linked to industrial discharge and sedimentation from farming.

Tasks

(a) Explain the effect of human activities on the ecosystem.

(b) Explain how these changes hinder Social-economic development.

(c) Suggest solutions to the challenges and benefits of implementing the solutions.

Part II

Answer only **one** *item from this part.*

Item 6

Sarah a 16 year old student, for several weeks experienced frequent urination, excessive thirst, tiredness, blurred vision and un explained weight loss. One morning at school, she fainted during assembly and was rushed to a nearby health centre. The heath worker suspected diabetes and tested Sarah's blood glucose level before and after she ate a meal. Her urine was also tested for glucose and proteins.

Table 1 shows Sarah's Blood glucose levels.

Time	Blood glucose levels mmol/*l*
Before meal	12.5
1 hour after meal	19.8
2 hours after meal	17.6
4 hours after meal	14.2

Urine test results showed absence of proteins and presence of glucose; and the normal fasting glucose level is 4-6 mmol/*l*.

Task

(a) (i) Using the information in the scenario and data in the table 1, explain why the health worker suspected diabetes.

(ii) Explain how diabetes affected the functioning of Sarah's body.

(b) Suggest suitable ways Sarah can manage her condition without being referred to another hospital and explain what may happen to Sarah if the condition is NOT managed properly.

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Item 7

Chemongesi is a talented long-distance runner, who smokes regularly. Their home is located near a swampy area. Over the past six months, Chemongesi started experiencing frequent illnesses. Medical tests later confirmed that he has developed heart and respiratory complications, alongside recurrent malaria infections. His Coach noticed a steady decline in his athletic performance and collected the following data;

Table 2: Chemongesi's monthly performance.

Month	Distance covered	Malaria episodes (number of times suffered)	Body Weight (Kg)	Lung function lung volume (cm^3)
January	6.5	0	60	420
February	6.2	1	59	400
March	5.8	2	57	370
April	5.1	2	55	340
May	4.3	3	53	310
June	3.5	3	50	280

Task

(a) Using the data provided, explain the decline in Chemongesi's sports performance.

(b) Suggest measures Chemongesi can take to manage his health condition and improve his- wellbeing.

END

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Candidate's Name: .....

Signature: ..... Personal No. .....

553/1
BIOLOGY
Paper 1
Jul / Aug 2026
2½ Hours

JINJA JOINT EXAMINATIONS BOARD

Uganda Certificate of Lower Secondary Education

MOCK EXAMINATIONS - JULY / AUGUST 2026

BIOLOGY

Paper 1

Theory

2 Hours 30 Minutes

INSTRUCTIONS TO CANDIDATES:

This paper consists of seven examination items. It has two sections; **A** and **B**.

Section **A** has **three** compulsory items.

Section **B** has two parts; **I** and **II**. Answer **one** item from each part.

Answer **five** items in all.

Any additional item(s) answered will not be scored.

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SECTION A

*Answer **all** the items in this section in the spaces provided.*

ITEM 1

A traditional herbalist collects herbal medicine from medicinal plants in a nearby forest. In the process of harvesting, he frequently plucks leaves and flowers, digs out roots, and strips bark from tree stems. Over time, the population of many medicinal plant species in the forest has greatly reduced. However, one particular species, which has numerous buds and a high capacity for regeneration, has continued to survive and remain abundant.

Task

a) Explain how the herbalist's harvesting methods affect the normal functioning of medicinal plants.

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b) Explain why one particular species has continued to survive despite the poor harvesting methods.

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ITEM 2

An old woman has been suffering from joint pain and frequent fractures. Her son developed visual and hearing impairment. When the son was diagnosed in hospital, the clinical laboratory report indicated that the auditory canals had a lot of wax and the ear drums and cochlea were damaged. The iris had less pigment and the ciliary muscles were weak. While in the hospital, the old woman was informed that her sickness was due to hormonal imbalance as a result of her old age. The woman is so worried about her health condition.

Task

1. Explain the cause of the son's impairments as indicated in the report.

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2. Explain what can be done by the old woman to manage her health condition.

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ITEM 3

A 15-year-old Senior Three homozygous normal girl became pregnant after engaging in a sexual relationship with a heterozygous normal man from a family with some albino members. The pregnancy has negatively affected her. She is also worried that the unborn baby may be albino because albinism exists in the man's family. Due to this and the challenges associated with the pregnancy, she is considering abortion. However, her friend has advised her against making that decision.

Task

a) Genetically show to the girl the possible phenotypes of the unborn child.

b) Explain the likely challenges that the girl could be experiencing due to her current condition.

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c) Explain the importance of the friend's advice to the girl.

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SECTION B

Answer two items from this section.

PART I

*Answer only **one** item from this part. Answers should be written in the answer booklets provided.*

ITEM 4

A farmer cleared a large area of natural vegetation to establish a crop field. He began growing maize intercropped with millet and, during the first three planting seasons, he obtained high yields. After several seasons of growing the same crops, the yields started to decline, and pest infestations became more severe. In response, the farmer began applying artificial fertilizers and broad-spectrum pesticides every season. Although yields improved temporarily, they later declined again. He also noticed that the fruit trees growing near the crop field produced fewer fruits than before.

Task

a) Explain how the activities of the farmer caused the observed challenges.
b) Advise the farmer on how he can sustainably ensure high yields.

ITEM 5

In the dry plains of a cattle corridor in Uganda, residents cut down trees for charcoal burning, burn grass during the dry season to control parasites and keep large herds of cattle. As a result, the hunters have observed a reduction in wild animals in the area, increased populations of rats in their homes and increased environmental temperature.

Task

a) Explain how the challenges in the area were brought about by the activities of the residents.
b) Why is it important for the residents to sustainably use the area?

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PART II

*Answer only **one** item from this part. Answers should be written in the answer booklets provided.*

ITEM 6

A five-year-old boy in primary one, suffered nutritional deficiencies which presented the following signs and symptoms; bow-shaped legs, bleeding gums, pot belly and a larger body size for his age. His Physical Education teacher realized that he is extremely slow during physical exercises but encouraged him to continue in order to improve his health. One day, during exercise, the teacher observed that the boy's heartbeat and breathing rate increased so much and he was complaining of muscle pain. She advised him to rest but the boy continued breathing fast for a while before returning to the normal rate.

Task

a) Explain the causes of the boy's health challenges.
b) Explain the body changes that occurred during and after exercise.

ITEM 7

A veterinary officer has been mixing his drugs while smoking cigarettes inside his poorly ventilated drug shop before going out to treat animals for his customers. He usually misses lunch since he spends the whole day attending to his various customers. Recently, he became ill. He was taken to hospital and the clinical laboratory report indicated that he has broken alveolar walls, very few cilia lining the air passages, wounds in the inner lining of the stomach and few villi in the ileum.

Task

a) Explain how the observations in the report led to the officer's ill health.
b) Apart from treatment, what advice can you give to him to improve his health?

END

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Candidate's Name: .....

Signature:.....

Random No.						Personal No.		

(Do not write your School Name or Number anywhere on this booklet.)

553/1
BIOLOGY
Paper 1
(Theory)
Jul./Aug. 2026
2½ hours

For Scorers' Use Only		
Total Weighted score		
Initials		

WAKISO-KAMPALA TEACHERS' ASSOCIATION (WAKATA)

WAKATA MOCK EXAMINATIONS 2026

Uganda Certificate of Education

BIOLOGY

(Theory)

Paper 1

2 hours 30 minutes

WAKATA 2026

INSTRUCTIONS TO CANDIDATES:

This paper consists of **two** *sections;* **A** *and* **B**. *It has* **seven** *examination items.*

Section **A** *has* **three compulsory** items.

Responses to Section **A must** *be written in in the spaces provided.*

Section B *has* **two** *parts;* **I** *and* **I**. *Respond to* **one** *item from each part.*

Responses to Section **B must** *be written in in the answer booklet(s) provided.*

Responses to each item in Section **B** *should start on a fresh page.*

Respond to **five** *items in all.*

Any additional items responded to will **not** *be scored.*

For Scorers' Use Only			
Item No.	**Basis code**	**Weighted score**	**Scorers' Initials**
1	PU		
	PA		
	CU		
	CA		
2	RU		
	RA		
____(4/5)	EU		
	EA		
____(6/7)	MU		
	MA		
Total Weighted Score			

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SECTION A

Respond to all items in this section in the spaces provided.

Item 1.

Sarah is a tomato farmer in her village. She transplanted tomato seedlings from a nursery bed to the main garden at the beginning of the rainy season. She noticed that seedlings transplanted late in the evening remained fresh and upright the following day. However, some seedlings transplanted during a hot afternoon wilted shortly after transplanting. On days with strong sunshine and dry winds, the leaves drooped and curled, while on cool cloudy days the plants appeared fresh and turgid. After a few days, most of the seedlings recovered and resumed normal growth.

Task:

(a) Explain how the environmental conditions affected the tomato seedlings after transplanting.

For Scorer's Use Only

(b) Suggest farming practices Sarah can use to reduce wilting of seedlings during transplanting.

For Scorer's Use Only

Basic code	PA
Score	

Item 2.

On his way home from school in the evening, Peter was walking through a bushy path when he suddenly stepped near a snake hidden in the grass. The snake moved quickly towards him, and Peter immediately jumped backward and moved away before he consciously thought about what had happened. His heart started beating faster, he began sweating, and his breathing rate increased. After a few seconds, he realized that his body had reacted automatically to protect him from danger.

Task:

(a) Explain to Peter how his body was able to respond quickly to the snake before he became fully aware of the danger.

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For Scorer's Use Only

Basic code	CU
Score	

(b) (i) Explain the importance of reflex actions to humans in everyday life.

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ii) Suggest ways people can reduce the risk of accidents caused by dangerous animals such as snakes in the community.

For Scorer's Use Only

Basic code	CA
Score	

Item 3.

A young couple went for medical screening before getting married. The results showed that the man had normal haemoglobin (AA) while the woman had sickle cell trait (AS). They ignored the advice given by the health workers and later got married. After some years, they had a child who frequently suffered from severe body weakness, repeated infections, joint pains, and required regular hospital care. The couple started blaming each other, with the father accusing the mother of causing the child's condition.

Task:

(a) (i) As a health genetic counsellor, explain to the couple the possible cause of their child's condition and how you would help them solve the conflict in their family.

CamScanner

(ii) Explain the possible genetic outcomes of their future children.

For Scorer's Use Only	
Basic code	**RU**
Score	

(b) Advise the couple on measures they can take to reduce the risk of having severely affected children.

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For Scorer's Use Only

Basic code	RA
Score	

SECTION B

Part I

Respond to only **one** *item from this part.*
Responses should be written in the answer booklet(s) provided.

Item 4.

In a hilly area, a farmer cleared vegetation on the slopes of the hill to create more land for farming to grow Irish potatoes. He cultivated up and down the slope and left the soil bare after harvesting. During the rainy season, heavy rainfall caused large amount of water to flow rapidly downhill.

Over time, small channels formed on the land and later developed into deep channels, much of the topsoil was washed away, in the subsequent seasons the Irish potatoes began to grow poorly, and nearby rivers became muddy and some of his land on the slopes of the hill became unusable. An agricultural officer visited the area and advised the farmer to adopt better land management practices to protect the soil to remain productive.

Task:

(a) Explain the challenges observed by the farmer.

(b) Suggest ways farmers can overcome such challenges.

Item 5.

A community living near a wetland depends on it for water, fishing, harvesting papyrus, and grazing animals. Due to increasing population, some farmers started clearing the wetland vegetation to create more land for growing crops.

They also began applying chemical fertilisers and pesticides which are washed into the wetland during heavy rains. Some residents dump domestic waste such as plastic containers, polythene bags, and sewage into the wetland. Over time, the wetland water became dirty, fish numbers reduced, some water plants disappeared, and flooding became more frequent in nearby villages.

Task:

(a) Explain to the community the possible environmental challenges they are likely to cause if they continue with these activities in the wetland.

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(b) Explain why the resources in the wetland should be conserved.

Part II

Respond to only **one** *item from this part.*
Responses should be written in the answer booklet(s) provided.

Item 6.

A group of Senior Four students visited a regional hospital during a biology study tour on human body processes. In the male medical ward, they met a 35-year-old patient. The doctor explained that the patient's condition had developed gradually over several years due to unhealthy lifestyle habits.

He frequently consumed highly salted and processed fast foods, rarely drank enough clean water, and often took medicines without medical prescription whenever he fell sick. He also consumed excessive amounts of alcohol and engaged in little or no physical exercise and used to smoke cigarettes daily.

Before admission, the patient showed clear signs of illness such as swollen feet and face, persistent tiredness, loss of appetite, chest pain and reduced urine output. Laboratory tes revealed high levels of urea and other nitrogenous wastes in his blood, indicating that his kidneys were not functioning properly.

WAKATA 2026

Task:

(a) (i) Describe the process involved in the removal of urea and other nitrogenous wastes from the body.

(ii) Explain the challenges the patient experienced before admission.

(b) Suggest ways the challenges experienced by the patient can be treated.

Item 7.

Sarah, a 28-year-old teacher, was involved in a motorcycle accident while travelling to work. During the accident, she suffered deep cuts on her legs and abdomen, resulting in heav bleeding. She also sustained injuries to her ribs and chest, making breathing difficult. By the time she reached the hospital, she was pale, weak, dizzy, and unconscious.
At the hospital, doctors immediately administered an intravenous glucose solution, transfused compatible blood, supplied oxygen through a face mask, and treated and dressed her wounds. After a few hours, Sarah regained consciousness, her breathing improved, and she gradually regained strength. However, further examination using a CT scan revealed that one of her lungs had been partially collapsed due to the chest injury.

Task:

(a) Explain how the motorcycle accident affected the functioning of Sarah's body.

(b) Describe how the medical interventions carried out by the doctors enabled Sarah to return to a near-normal state.

END

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Candidate's Name: .....

Signature: .....

Random No.						Personal No.		

(Do not write your School Name or Number anywhere on this booklet.)

545/1
CHEMISTRY
Paper 1
(Theory)
Jul./Aug. 2026
2 hours

For Scorers' Use Only		
Total Weighted score		
Initials		

WAKISO-KAMPALA TEACHERS' ASSOCIATION (WAKATA)

WAKATA MOCK EXAMINATIONS 2026

Uganda Certificate of Education

CHEMISTRY

(Theory)

Paper 1

2 hours

WAKATA 2026

INSTRUCTIONS TO CANDIDATES:

This examination paper consists of **two** *sections;* **A** *and* **B**.

It has **seven** *examination items.*

Section **A** *has* **two compulsory** items. *Responses to Section* **A must** *be written in in the spaces provided.*

Section B *has* **two** *parts;* **I** *and* **I**. *Respond to* **one** *item from each part.*

Responses to Section **B must** *be written in in the answer booklet(s) provided.*

Respond to **four** *items in all.*

Responses to each item in Section **B** *should start on a fresh page.*

Any additional item(s) responded to will **not** *be scored.*

For Scorers' Use Only			
Item No.	Basis code	Weighted score	Scorers' Initials
1	T		
	D		
	E		
2	C		
	X		
	I		
____(3/4)	P		
	S		
	B		
____(5/6)	N		
	M		
	Be		
Total Weighted Score			

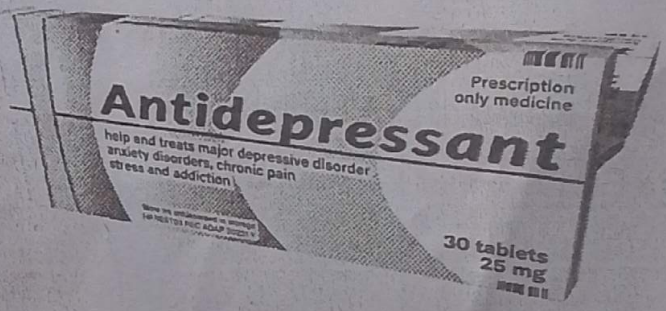
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SECTION A

Respond to **all** *items in this section in the spaces provided.*

Item 1.

Dr Musa is the head of the cancer ward in a hospital located in his home town. Due to the nature of his work, he often feels very tired, depressed, and experiences high levels of tension and anxiety. He was advised by a close friend to use natural remedies, but instead he visited his personal doctor, who prescribed tablets to help reduce his condition.



Task:

As a chemistry learner:

(a) State the category and the function of the product used.

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For Scorer's Use Only

Basis Code	T
Score	

(b) Describe the side effects associated with prolonged use of the product.

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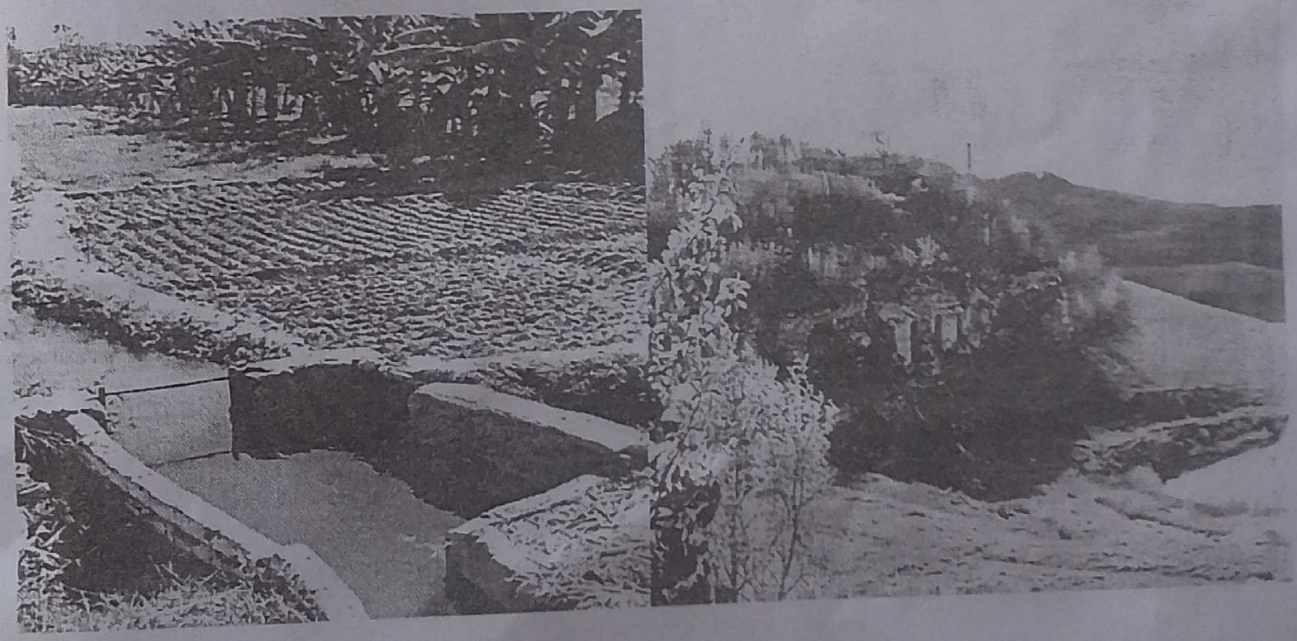
Part II

Respond to only **one** *item from this part in the answer booklet(s) provided*

Item 5.

Mrs. Mwasa is a farmer who specialises in large-scale crop husbandry. She grows her crops along the only river from which the rest of the community obtains fresh water. As a result of her continued farming practices, local residents have begun complaining that the river water is no longer clean for domestic use or for their animals.

The Local Council chairperson has organised a community meeting to address the concerns raised by the residents.



Task:

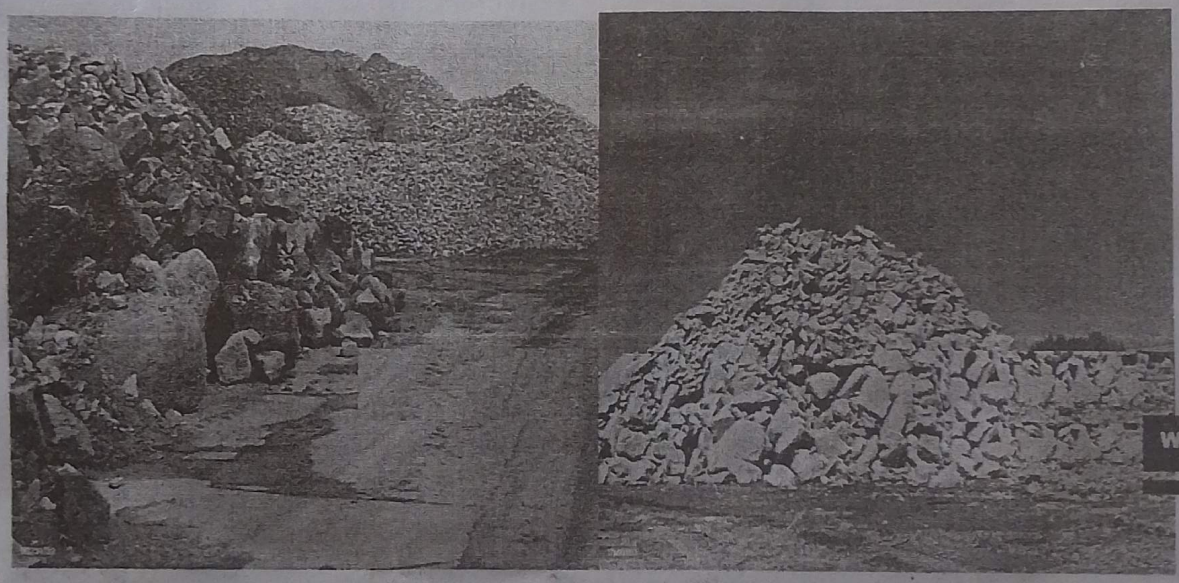
As a chemistry learner, write a speech that you will present to the chairperson to help him or her address the community.

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Item 6.

During the school holidays, Eva usually visits her father in their home village. Her father tells her that he earns her school fees from stone quarrying and sells the gravel to support the family. However, because of this activity, local residents have begun expressing concern about its long-term effects on the environment.

The local environmental authorities have organised a sensitisation workshop.



WAKATA 2026

Task:

As a chemistry student, prepare a write-up that you will present to the community.

CamScanner

Name: ..... Centre/Index No. .....

School: ..... Signature: .....

545/2
CHEMISTRY
Paper 2
(Practical)
July/August 2026
2 hours

For Scorers' use Only

Total Weighted Score		
Initials		

WAKISSHA JOINT MOCK EXAMINATIONS
Uganda Certificate of Education

FOR SCORERS' USE ONLY

Basis Code	Weighted Score	Scorers' Initials
A		
H		
V		
P		
R		
D		
Dr		
I		
C		
Total Weighted Score		

INSTRUCTIONS TO CANDIDATES:

- *This paper consists of* **one** *compulsory examination item.*
  Responses to this item are to be written in the spaces provided in this booklet. Use blue or black ink.
- *All working must be clearly shown.*
- *Mathematical tables and silent non programmable scientific calculators may be use; You are not allowed to use reference books.*
- *Candidates are advised to carefully read the item, make sure you have all the apparatus and chemicals you may need and then plan appropriately before starting.*

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Turn Over

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Item 1

Koile, one of the progressive farmers in Eastern Uganda uses lime water to lower the acidity of soils in his garden so as to ensure proper growth of his vegetables. He has got two samples of lime water from two suppliers X and Y but he is not sure of which one is more effective. A local agriculture extension worker advises that the lime water which is more alkaline (Better at neutralizing the acid in the soil), requires a larger volume of dilute hydrochloric acid solution to reach neutralization point when titrated with the acid. As a member of the science club in your school you have been requested to assist Koile by pipetting fixed volumes of the lime water solution samples separately and titrating them with dilute hydrochloric acid solution using phenolphthalein indicator;

You have been provided with the following:

- DA2 which is lime water solution from supplier X;
- DA3 which is lime water solution from supplier Y;
- DA1 which is dilute hydrochloric solution;
- Phenolpthalein indicator solution.
- Some laboratory apparatus.

Task

As a chemistry learner,

(a) Design an experiment you will carry out to help Koile. (Your design should include the; aim, hypothesis, variables, apparatus and materials, procedure, risks and their mitigations).

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For Scorers' use only

Basis code	A	H	V	P	R
Score					

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(b) Carry out the experiment and record your results.

For Scorers' use only

Basis code	D	Dr.
Score		

(c) Analyse your results and advise Koile accordingly.

For Scorers' use only

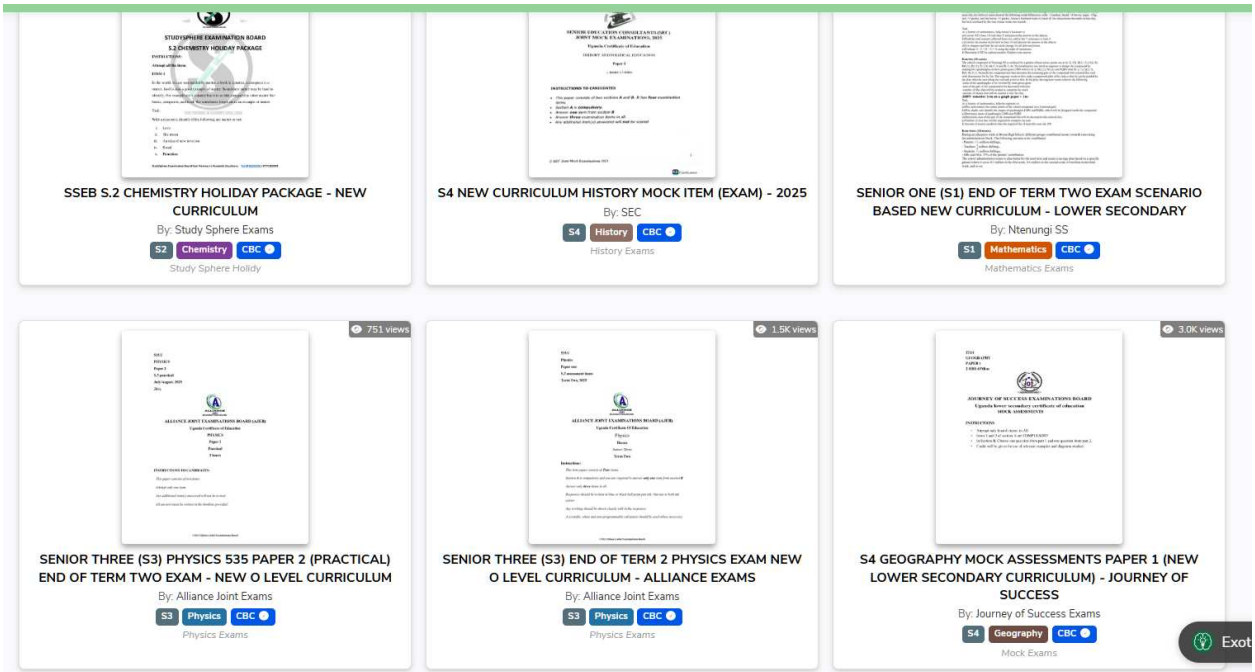
Basis code	I	C
Score		

END

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